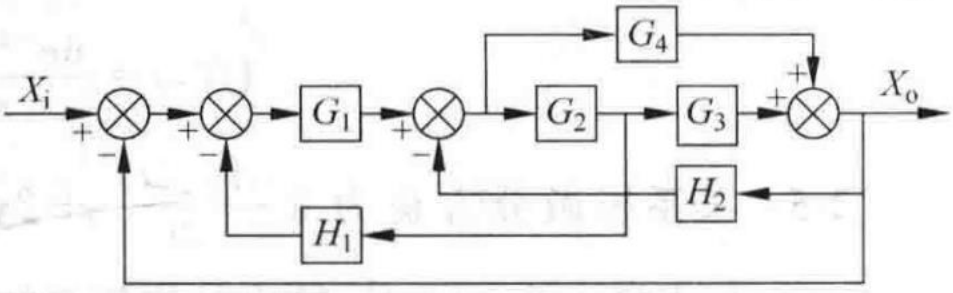
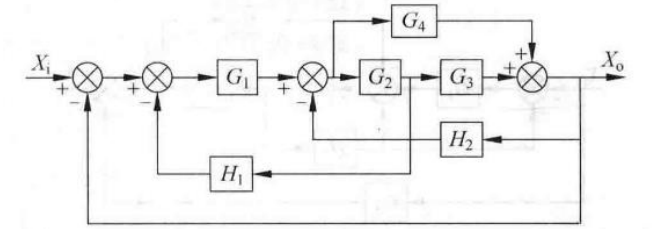
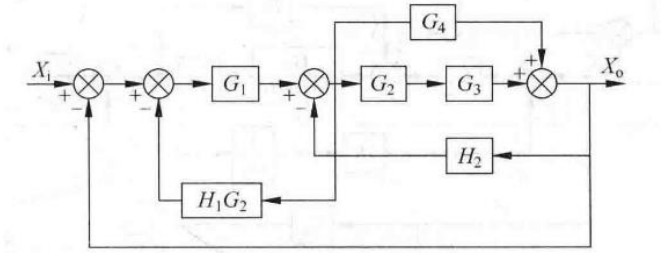
2-6 化简图 2.2(a)～(d)所示的各方块图，并确定其传递函数。

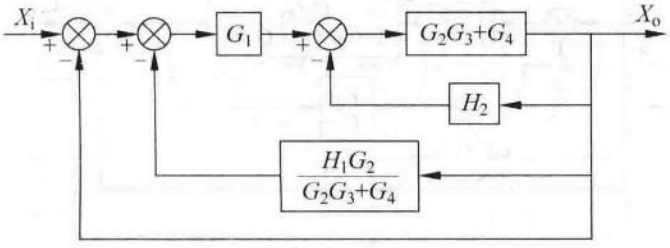


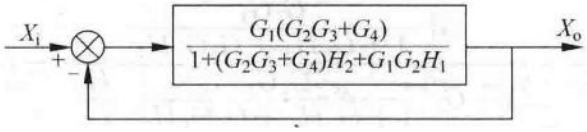
(b)

解：









$$\frac{X_o}{X_i}=\frac{\dfrac{G_1(G_2G_3+G_4)}{1+(G_2G_3+G_4)H_2+G_1G_2H_1}}{1+\dfrac{G_1(G_2G_3+G_4)}{1+(G_2G_3+G_4)H_2+G_1G_2H_1}}$$

$$=\frac{G_1(G_2G_3+G_4)}{1+(G_2G_3+G_4)(G_1+H_2)+G_1G_2H_1}$$

2-7 对于图 2.7 所示的系统：

(1) 求 $X_o(s)$和 $X_{i1}(s)$之间的闭环传递函数；

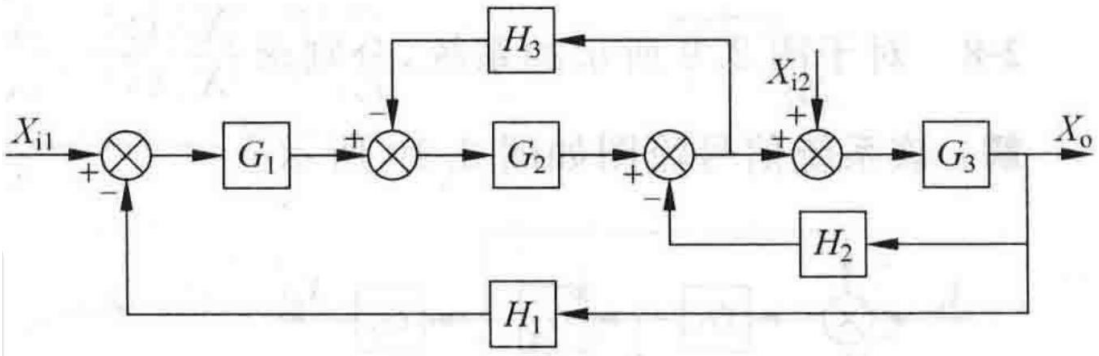


图 2.7

解：

通过方块图变换，该系统与图 2.8 所示系统等价。

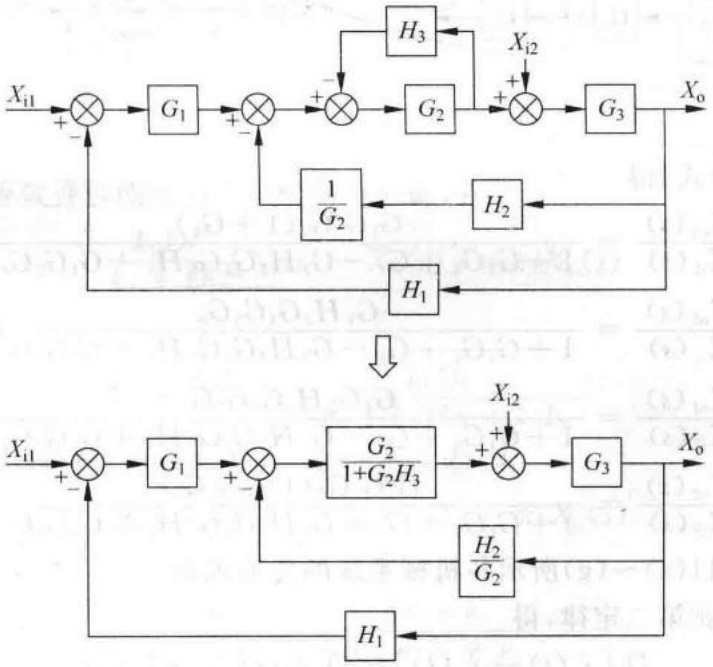


图 2.8

根据化简后的方块图，得

$$(1)\ \frac{X_o(s)}{X_{i1}(s)}=\frac{G_1\dfrac{G_2}{1+G_2H_3}G_3}{1+\dfrac{G_2}{1+G_2H_3}G_3\dfrac{H_2}{G_2}+G_1\dfrac{G_2}{1+G_2H_3}G_3H_1}$$

$$=\frac{G_1G_2G_3}{1+G_2H_3+G_3H_2+G_1G_2G_3H_1}$$

2-8 对于图 2.9 所示的系统，分别求 $\dfrac{X_{o1}(s)}{X_{i1}(s)}$、$\dfrac{X_{o2}(s)}{X_{i2}(s)}$、$\dfrac{X_{o1}(s)}{X_{i2}(s)}$、$\dfrac{X_{o2}(s)}{X_{i1}(s)}$。

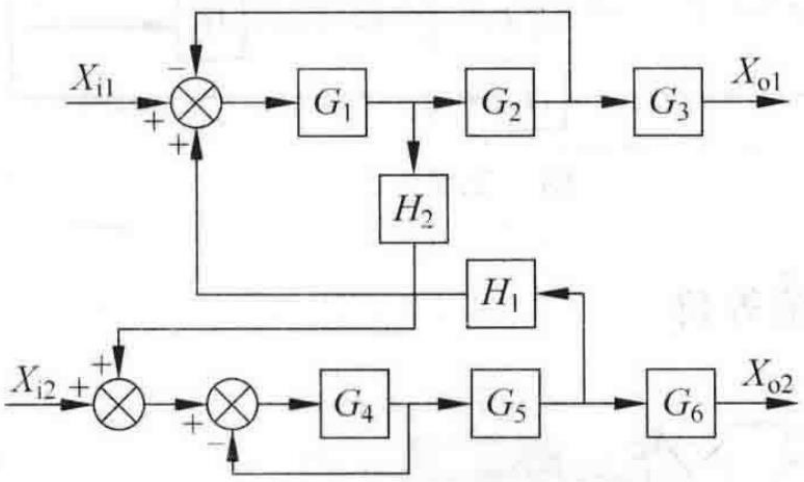


图 2.9

解：

该系统信号流图如图 2.10 所示。

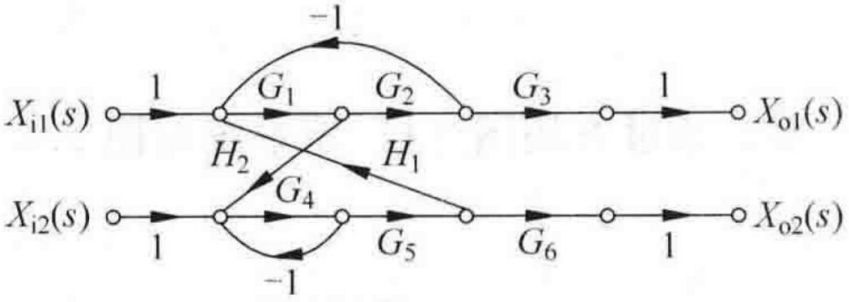


图 2.10

根据梅逊增益公式，得

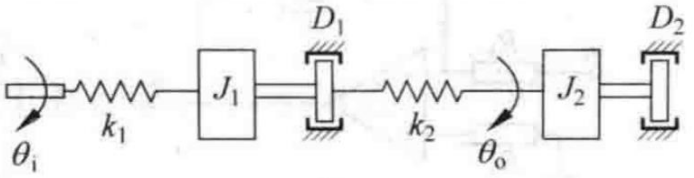
$$\frac{X_{o1}(s)}{X_{i1}(s)}=\frac{G_1G_2G_3(1+G_4)}{1+G_1G_2+G_4-G_1H_2G_4G_5H_1+G_1G_2G_4}$$

$$\frac{X_{o2}(s)}{X_{i1}(s)}=\frac{G_1H_2G_4G_5G_6}{1+G_1G_2+G_4-G_1H_2G_4G_5H_1+G_1G_2G_4}$$

$$\frac{X_{o1}(s)}{X_{i2}(s)}=\frac{G_4G_5H_1G_1G_2G_3}{1+G_1G_2+G_4-G_1H_2G_4G_5H_1+G_1G_2G_4}$$

$$\frac{X_{o2}(s)}{X_{i2}(s)}=\frac{G_4G_5G_6(1+G_1G_2)}{1+G_1G_2+G_4-G_1H_2G_4G_5H_1+G_1G_2G_4}$$

2-12 试求图 2.15(a)、(b)所示机械系统的传递函数。



(b)

解：

(b) 系统可抽象成如图 2.16 所示的方块图。

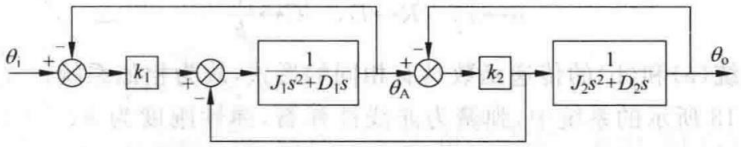


图 2.16

根据方块图可得

$$\frac{\Theta_o(s)}{\Theta_i(s)}=\frac{\dfrac{k_1}{J_1s^2+D_1s}\cdot\dfrac{k_2}{J_2s^2+D_2s}}{1+\dfrac{k_1}{J_1s^2+D_1s}+\dfrac{k_2}{J_2s^2+D_2s}+\dfrac{k_2}{J_1s^2+D_1s}+\dfrac{k_1}{J_1s^2+D_1s}\cdot\dfrac{k_2}{J_2s^2+D_2s}}$$

$$=\frac{1}{\dfrac{J_1J_2}{k_1k_2}s^4+\dfrac{J_1D_2+J_2D_1}{k_1k_2}s^3+\left(\dfrac{J_1}{k_1}+\dfrac{J_2}{k_2}+\dfrac{J_2}{k_1}+\dfrac{D_1D_2}{k_1k_2}\right)s^2+\left(\dfrac{D_1}{k_1}+\dfrac{D_2}{k_2}+\dfrac{D_2}{k_1}\right)s+1}$$

2-19 某机械系统如图 2.24 所示。其中，M_1 和 M_2 为质量块的质量，D_1、D_2 和 D_3 分别为质量块 M_1、质量块 M_2 和基础之间，质量块之间的黏性阻尼系数。$f_i(t)$为输入外力，$y_1(t)$和 $y_2(t)$分别为两质量块 M_1 和 M_2 的位移。试求 $G_1(s)=\dfrac{Y_1(s)}{F_i(s)}$和$G_2(s)=\dfrac{Y_2(s)}{F_i(s)}$。

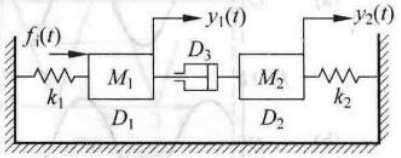


图 2.24

解：该系统可抽象成如图 2.25 所示的方块图。

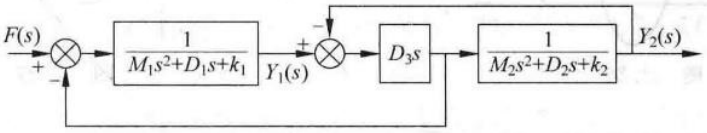


图 2.25

由方块图可得

$$G_1(s)=\frac{M_2s^2+(D_2+D_3)s+k_2}{M_1M_2s^4+(M_1D_2+M_2D_1+M_1D_3+M_2D_3)s^3+(M_1k_2+M_2k_1+D_1D_2+D_1D_3+D_2D_3)s^2+(D_1k_2+D_2k_1+D_3k_1+D_3k_2)s+k_1k_2}$$

$$G_2(s)=\frac{D_3s}{M_1M_2s^4+(M_1D_2+M_2D_1+M_1D_3+M_2D_3)s^3+(M_1k_2+M_2k_1+D_1D_2+D_1D_3+D_2D_3)s^2+(D_1k_2+D_2k_1+D_3k_1+D_3k_2)s+k_1k_2}$$